Scalability Analysis of Multi-Objective Metaheuristics for Assembly Line Balancing under Soft Zoning Constraints

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***Abstract*—This study addresses the Multi-Objective Assembly Line Balancing Problem with Soft Zoning Constraints (MOALBP-SZ) to minimize both the number of workstations and the total operational costs, which include penalties for zoning rule violations. Traditional exact solvers frequently suffer from exponential growth of the search space, leading to severe computational deadlocks in large-scale industrial scenarios. To overcome this limitation, the research formulates a Mixed-Integer Linear Programming (MILP) model and comprehensively evaluates the scalability of six algorithmic approaches: the Exact Method, Pure VNS, Hybrid VNS, NSGA-II, MOEA/D, and Multi-Objective Variable Neighborhood Search (MO-VNS). Computational experiments were conducted across small, medium, and large benchmark datasets. The empirical results demonstrate that while the Exact Method and Hybrid VNS operate efficiently on small instances, the Exact Method experiences a complete computational collapse on large-scale datasets. In contrast, MO-VNS exhibits exceptional computational speed and a balance of Multi-Objective, making it the most practical choice for Small and Medium-sized Enterprises requiring real-time responsiveness. For large-scale enterprises with highly complex constraints, NSGA-II proves to be the most robust algorithm, successfully maintaining solution diversity and identifying high-quality Pareto optimal alternatives within seconds.**

***Index Terms*—Multi-Objective Optimization, Assembly Line Balancing, Soft Zoning Constraints, Metaheuristics**

I. Introduction

A. Background and Significance of the Problem

In the highly competitive era of industrial manufacturing, maximizing resource management efficiency is a critical factor for organizational success. The Assembly Line Balancing Problem (ALBP) remains a classic challenge in Operations Research, with the primary objective of allocating tasks to workstations to achieve balance, minimize idle time, and maximize production rates. [1]

However, real-world manufacturing assembly lines are often characterized by complex conditions. [2] One of the most challenging is the Zoning Constraint, which dictates that certain pairs of tasks cannot be assigned to the same workstation. Examples include high-heat generating tasks that must be separated from high-precision measurement tasks. Traditionally, mathematical models have treated these as Hard Constraints. In practice, when production problems are large-scale and highly complex, enforcing such absolute rules often leads to Infeasible Solutions.

To address this, the concept of Soft Constraints has been introduced. This approach allows for the violation of zoning rules when necessary, but at the cost of a Penalty Cost. Implementing soft constraints transforms the ALBP into a Multi-Objective Optimization problem, where decision-makers must balance the reduction of workstations (Z_1) against the minimization of total costs, including penalty costs (Z_2). [3]

B. Research Gap

A review of existing literature indicates that most research focuses either on Single-Objective ALBP or ignores zoning constraints in multi-objective contexts, often relying on Hard Constraints. Furthermore, finding an optimal solution using Exact Methods, such as Mixed-Integer Linear Programming (MILP) [4], frequently encountered strict limitations due to the exponential growth of the search space. These methods often fail to provide solutions within an acceptable time frame as the problem sizes scale to industrial levels. While various meta-

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heuristics have been applied, a systematic comparative study between Exact Methods and Multi-Objective Metaheuristics across different problem scales such as zoning constraints and different modes of operation, remains a significant gap in the current body of knowledge. [5]

C. Research Objectives

This research aims to bridge these gaps with the following primary objectives:

- To develop a mathematical model for the Multi-Objective Assembly Line Balancing Problem that incorporates Soft Zoning Constraints.
- To develop and apply five algorithmic approaches: Exact Method, Hybrid VNS, Pure VNS, and high-level Multi-Objective Metaheuristics (NSGA-II, MOEA/D, and MO-VNS).
- To evaluate and compare the performance of these five approaches using international performance metrics, including Hypervolume Ratio (HVR), Additive Epsilon Indicator, and Computational Time across various benchmark datasets.

D. Scope and Expected Benefits

This research conducts evaluations on standard datasets categorized into three groups based on complexity: Group A (Small), Group B (Medium), and Group C (Large). The findings are expected to provide theoretical evidence regarding the limitations of Exact Methods and offer practical guidelines for the industrial sector in selecting the most appropriate algorithms based on the specific scale and context of their assembly lines.

II. Related Work

The Assembly Line Balancing Problem is a fundamental combinatorial optimization challenge aimed at assigning tasks to workstations efficiently. In practical manufacturing environments, spatial restrictions such as zoning constraints frequently occur, preventing incompatible tasks from sharing the same workstation. [6] While traditional mathematical models enforce these restrictions strictly, doing so often leads to infeasible solutions in highly constrained scenarios. Recent approaches relax these spatial restrictions into soft constraints, permitting violations at a penalty cost. This transformation inherently turns the traditional balancing model into a multi objective optimization problem.

To solve these complex formulations, Mixed Integer Linear Programming is conventionally employed to secure global optimal solutions. However, exact methods fundamentally suffer from the exponential growth of the search space. [4] As the problem size expands through additional execution modes and soft zoning variables, the search space grows exponentially. [7], [8] Consequently, commercial exact solvers become computationally prohibitive for medium and large scale industrial instances, highlighting an urgent necessity for efficient approximation algorithms.

In dealing with conflicting objectives, such as minimizing the number of workstations versus minimizing the total penalty costs, a single optimal solution cannot exist. The objective shifts toward approximating the true Pareto Front, which represents a set of optimal alternative solutions based on Pareto dominance. The quality of these approximated solutions must be rigorously evaluated using quantitative performance metrics. Researchers typically rely on the Hypervolume Ratio to measure solution diversity and objective space coverage, alongside the Additive Epsilon Indicator to assess convergence toward the ideal reference front. [9]–[11]

To overcome the severe computational bottlenecks of exact mathematical models, the academic community has widely adopted multi objective metaheuristics. Variable Neighborhood Search (VNS) provides rapid convergence through systematic shaking and local search mechanisms, proving highly efficient for rapid decision making in smaller instances. [12] For larger and more complex search spaces, population based algorithms are often preferred. The Non dominated Sorting Genetic Algorithm II (NSGA-II) utilizes fast sorting and crowding distance evaluation to robustly maintain solution diversity. [13] Alternatively, the Multi Objective Evolutionary Algorithm based on Decomposition (MOEA/D) divides the main problem into multiple scalar subproblems using weight vectors, demonstrating exceptional convergence capabilities when dealing with strongly conflicting objectives and high-level Multi-Objective Variable Neighborhood Search (MO-VNS) [14], [15]

III. Methodology

This section presents the research methodology developed to solve the Multi-Objective Assembly Line Balancing Problem with Soft Zoning Constraints (MOALBP-SZ). The approach is fundamentally divided into two main components: the mathematical formulation of the problem and the proposed solution algorithms.

A. Mathematical Formulation

To accommodate the complexities of real-world manufacturing environments, the traditional hard constraints regarding spatial restrictions are relaxed into soft constraints. Incompatible tasks are permitted to be assigned to the same workstation; however, such violations incur a substantial penalty cost. The Mixed-Integer Linear Programming (MILP) model is formulated as follows:

1) Indices and Sets:

- I: Set of all tasks, where $i, j \in I$.
- K: Set of available workstations, where $k \in K$.
- M_i: Set of execution modes for task i, where $m \in M_i$.
- P: Set of precedence relations (i, j) where task i must precede task j.
- Z: Set of incompatible task pairs (i, j) subjected to zoning constraints.

2) Parameters:

- C: Maximum allowable cycle time.
- t_{im}: Processing time of task i under execution mode m.
- c_{im}: Execution cost of task i under mode m.
- $Penalty$: A significantly large cost $(100,000)$ imposed for violating a zoning constraint.

3) Decision Variables:

- $x_{ikm} \in \{0,1\}$: 1 if task i is assigned to workstation k using mode m; 0 otherwise.
- $y_k \in \{0,1\}$: 1 if workstation k is utilized; 0 otherwise.
- $v_{ijk} \in \{0,1\}$: 1 if incompatible tasks i and j are assigned to the same workstation k; 0 otherwise.

4) Objective Functions: The proposed model aims to minimize two conflicting objectives simultaneously:

$$\text{Minimize } Z_1 = \sum_{k \in K} y_k \tag{1}$$

$$\text{Minimize } Z_2 = \sum_{i \in I} \sum_{k \in K} \sum_{m \in M_i} c_{im} x_{ikm} + \sum_{(i,j) \in Z} \sum_{k \in K} Penalty \cdot v_{ijk} \tag{2}$$

Equation (1) minimizes the total number of utilized workstations, while Equation (2) minimizes the total operational cost, including the execution costs and the zoning violation penalties.

5) Constraints:

$$\sum_{k \in K} \sum_{m \in M_i} x_{ikm} = 1 \quad \forall i \in I \tag{3}$$

$$\sum_{i \in I} \sum_{m \in M_i} t_{im} x_{ikm} \leq C \cdot y_k \quad \forall k \in K \tag{4}$$

$$\sum_{k \in K} \sum_{m \in M_i} k \cdot x_{ikm} \leq \sum_{k \in K} \sum_{m \in M_j} k \cdot x_{jkm} \quad \forall (i,j) \in P \tag{5}$$

$$\sum_{m \in M_i} x_{ikm} + \sum_{m \in M_j} x_{jkm} - v_{ijk} \leq 1 \quad \forall (i,j) \in Z, \forall k \in K \tag{6}$$

$$y_k \geq y_{k+1} \quad \forall k \in \{1, \ldots, |K| - 1\} \tag{7}$$

The constraint (3) ensures that each task is assigned to exactly one workstation with a specific mode. The constraint (4) guarantees that the total processing time in any workstation does not exceed the cycle time. The constraint (5) strictly enforces the precedence relations. The constraint (6) regulates the soft zoning mechanism, forcing v_{ijk} to 1 if incompatible tasks share a workstation. Finally, The constraint (7) ensures that the workstations are opened consecutively.

B. Data Structure

To evaluate the performance of the proposed algorithms in solving the multi-objective assembly line balancing problem, this study utilizes the well-known benchmark datasets originally developed by Otto as the foundational baseline. [16] However, because these original datasets were exclusively designed for the simple assembly line balancing problem, they strictly consider deterministic task times without reflecting the multifaceted constraints of modern manufacturing. To bridge this gap, we systematically extended these baseline instances into generalized assembly line balancing problems, thereby simulating the actual complexities encountered in real-world industrial environments. [17] The primary motivation behind these structural modifications is the creation of exceptionally hard benchmark instances. This approach effectively prevents the data saturation issues often encountered when advanced metaheuristics easily solve basic problems, thereby allowing us to truly stress-test the capabilities of algorithms like NSGA-II, MOEA/D, and MO-VNS. Transitioning these spatial rules into a soft constraint framework profoundly impacts the dual-objective mathematical model. The dense web of incompatible and linked tasks severely restricts task grouping flexibility, which naturally drives up the required number of workstations (Z_1). Simultaneously, the algorithms are forced into a constant state of trade-off evaluation for the second objective (Z_2). The system must intelligently navigate between the risk of violating zoning rules, which accumulates heavy penalty costs, and selecting the optimal mix of fast and slow execution modes to keep the total operational cost minimal, all while strictly adhering to the predetermined cycle time limit.

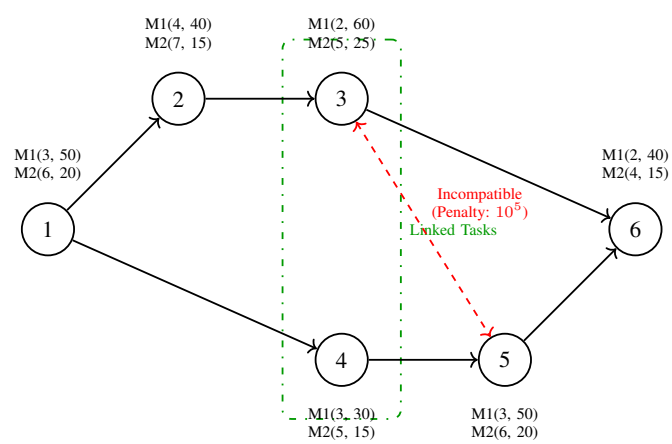

Fig. 1. Precedence graph illustrating multi-mode processing alternatives (Time, Cost), soft zoning constraint (incompatible tasks), and hard zoning constraint (linked tasks).

Fig. 1 illustrates the structural complexity of the proposed Multi-Objective Assembly Line Balancing Problem with Soft Zoning (MOALBP-SZ) using a representative precedence graph. Each circular node represents an assembly task and is explicitly annotated with its available execution modes, displaying the inherent trade-off between processing time and execution cost. The solid directed arrows establish the strict chronological precedence among the tasks. Furthermore, the graph highlights two critical spatial constraints integrated into the problem. The green bounding box denotes a hard zoning constraint (linked tasks), strictly mandating that the enclosed tasks (e.g., tasks 3 and 4) must be assigned to the exact same workstation. Conversely, the red dashed line represents a soft zoning constraint (incompatible tasks) between conflicting operations (e.g., tasks 3 and 5). If an algorithm assigns these

incompatible tasks to the same workstation, a substantial penalty cost (e.g., 10^5) is heavily accumulated into the total operational cost objective (Z_2). This comprehensive schematic effectively demonstrates the intricate decision-making process required by the algorithms to intelligently balance cycle time restrictions, execution costs, and spatial configurations.

C. Proposed Solution Approaches

To comprehensively evaluate the performance landscape of the MOALBP-SZ, five distinct algorithmic approaches are proposed and compared:

To address the inherent complexity of soft zoning constraints and multi-mode execution within the MOALBP-SZ, a standardized and robust decoding mechanism is imperative. Consequently, all metaheuristic algorithms evaluated in this study utilize a shared decoding scheme, as detailed in Algorithm 1. This systematic procedure transforms continuous or discrete task sequences into feasible assembly line configurations. Concurrently, it evaluates the objective functions by calculating the total number of utilized workstations (Z_1) and dynamically accumulating the total operational cost (Z_2), which strictly incorporates heavy penalty costs for any spatial zoning violations as shown in Fig. 2

Algorithm 1: Decoding Scheme for Soft Zoning Constraints

Input : Task sequence S, Cycle time limit C, Execution costs c_{im}, Zoning constraints matrix Z, Penalty cost
Output: Objective values: Number of workstations (Z_1), Total operational cost (Z_2)
$Z_1 \leftarrow 1$ Initialize the first workstation $Z_2 \leftarrow 0$ Initialize total operational cost $ST \leftarrow 0$ Initialize current station processing time $CurrentStationTasks \leftarrow \emptyset$;
foreach *task i and its selected execution mode m in sequence S* **do**
  1. Check Cycle Time Constraint **if** $ST + t_{im} > C$ **then**
    $Z_1 \leftarrow Z_1 + 1$ Open a new workstation $ST \leftarrow 0$ Reset station time $CurrentStationTasks \leftarrow \emptyset$ Clear tasks for the new station
  end
  2. Assign Task and Accumulate Operational Cost $ST \leftarrow ST + t_{im}$;
  $Z_2 \leftarrow Z_2 + c_{im}$ Add mode-specific execution cost
  3. Evaluate Soft Zoning Constraints
end
foreach *task $j \in CurrentStationTasks$* **do**
  if *pair $(i, j) \in Z$* (Incompatible Tasks) **then**
    $Z_2 \leftarrow Z_2 + Penalty$ Apply heavy penalty for zoning violation
  end
end
$CurrentStationTasks \leftarrow CurrentStationTasks \cup \{i\}$;
return Pareto Archive $\mathcal{A}$;

Fig. 2. Decoding Scheme for Soft Zoning Constraints and Multi-Mode Execution

1) Exact Method: A Mixed-Integer Linear Programming (MILP) solver (IBM ILOG CPLEX) is utilized to find the global optimum. This serves as the true reference benchmark for small-sized instances but is susceptible to the exponential growth of the search space as the problem size expands.

2) Single-Objective Metaheuristics: Initially, this research intended to propose a powerful matheuristic framework by integrating Variable Neighborhood Search (VNS) with an exact Constraint Programming (CP) solver. The core rationale was to leverage the CP solver during the decoding phase to rigorously search the local neighborhood for optimal task assignments that completely avoid or strictly minimize zoning penalties. While this hybrid architecture demonstrates exceptional accuracy in small-scale scenarios, empirical observations revealed a critical limitation. The CP solver introduces a severe computational bottleneck. As the problem dimensions expand to large-scale industrial instances, the exact nature of the solver becomes susceptible to the exponential growth of the search space, leading to extended processing times and computational deadlocks.

3) Multi-Objective Metaheuristics: To capture the Pareto Front representing optimal trade-offs, three advanced evolutionary algorithms are applied:

- NSGA-II: Employs fast non-dominated sorting and crowding distance mechanisms to maintain diversity within the complex search space, ensuring robust performance in large-scale instances.
- MOEA/D: Decomposes the multi-objective problem into several scalar optimization subproblems using weight vectors, facilitating rapid convergence toward the ideal reference point.
- MO-VNS: The proposed algorithm that adapts the shaking and local search mechanisms of traditional VNS into a multi-objective framework, offering a highly competitive balance between solution quality and computational speed.

These algorithms are explicitly designed to approximate the true Pareto Front by maintaining solution diversity and avoiding exhaustive exact evaluations. Experimental evaluations confirm that as the problem scale expands significantly, these pure metaheuristic approaches exhibit superior computational efficiency and robust scalability, successfully providing decision-makers with diverse, high-quality trade-off alternatives within seconds.

4) Parameter Tuning via Taguchi Method: The performance of metaheuristic algorithms is highly dependent on the configuration of their control parameters. To systematically determine the optimal parameter settings while minimizing the required number of computational experiments, the Taguchi method was employed as the design of experiments (DOE) approach. For each of the three multi-objective algorithms, an $L_9(3^4)$ orthogonal array was utilized, evaluating four key control factors at three distinct levels (low, medium, and high). [18]

The calibrated factors for each algorithm were defined as follows:

- NSGA-II: Population size (30, 50, 100), maximum generations (50, 100, 200), crossover rate (0.8, 0.9, 1.0), and mutation rate (0.05, 0.1, 0.2).
- MOEA/D: Population size (30, 50, 100), maximum generations (50, 100, 200), neighborhood size (T) (5, 10, 15), and mutation rate (0.05, 0.1, 0.2).

- MO-VNS: Maximum consecutive non-improving iterations (30, 50, 100), initial archive size (10, 20, 50), shaking strength (1, 2, 3), and local search limit (50, 100, 200).

The Signal-to-Noise (S/N) ratio was calculated for each parameter combination to evaluate the robustness and quality of the generated Pareto Fronts. The optimal parameter levels identified during this calibration phase were subsequently adopted for all comparative experiments to ensure a fair and rigorous performance assessment.

D. Performance Evaluation Metrics

To quantitatively assess the performance of the approximation algorithms against the exact method, two standard multi-objective metrics are utilized:

- Hypervolume Ratio (HVR): Evaluates the diversity and the extent of the objective space covered by the approximated Pareto Front. A value closer to 1.0 indicates superior performance.
- Additive Epsilon Indicator (I_ϵ): Measures the minimum translation distance required for an approximated front to completely dominate the true reference front. A value closer to 0.0 reflects better convergence.

IV. Experimental Study

This section presents the performance evaluation of the algorithmic approaches. Namely, the Exact Method (CPLEX), Hybrid VNS, Pure VNS, MOEA/D, NSGA-II, and MO-VNS. In solving the Multi-Objective Assembly Line Balancing Problem with Soft Zoning Constraints (MOALBP-SZ) To systematically analyze the impact of search space expansion on solution quality and computational time, the experimental results are categorized into three distinct problem sizes: small instances (Group A), medium instances (Group B), and large instances (Group C). [16]

A. Experimental Setup and Metrics

All computational experiments were conducted on the modified benchmark instances. The algorithmic performances were systematically recorded and comprehensively compared using three primary evaluation metrics:

- Hypervolume Ratio (HVR): This metric is utilized to measure the diversity and the extent of objective space coverage provided by the generated solution sets relative to the reference Pareto front.
- Additive Epsilon Indicator (I_ϵ): This indicator is employed to evaluate the convergence capability of the algorithms, measuring the distance required for the approximated front to reach the ideal solutions.
- Computational Time: Measured in seconds, this metric evaluates the computational efficiency and the practical suitability of the algorithms for real-world industrial implementation.

B. Hard vs. Soft Constraints Analysis

Prior to evaluating the main algorithms, a preliminary study was conducted using the Hybrid VNS to determine the optimal framework for zoning constraints by comparing hard and soft constraint models. The results indicate that in small and medium instances (Groups A and B), both models performed equally well, achieving a 100% feasibility rate with comparable computational times.

However, in the highly complex search spaces of large instances (Group C), the hard constraints model encountered severe computational deadlocks, with the feasibility rate plummeting to 0% in several cases. Conversely, relaxing these spatial rules into soft constraints that penalize rather than strictly prohibit violations successfully restored the feasibility rate to as high as 80%. Although this approach inherently increases total penalty costs, it provides viable, actionable production alternatives instead of computational failures. Consequently, the soft constraints framework was adopted as the baseline for all subsequent multi-objective experiments. The rationale for adopting the soft constraints framework, as well as the specific metrics chosen for this comparison, requires crucial clarification. In highly constrained, large-scale instances, enforcing strict hard constraints frequently leads to complete computational deadlocks, meaning the algorithm fails to identify any viable assembly line configuration. Transitioning to a soft constraints framework introduces essential flexibility by penalizing zoning violations with a substantial cost rather than strictly prohibiting them. This relaxation ensures that the system can consistently generate actionable production alternatives even in extreme scenarios.

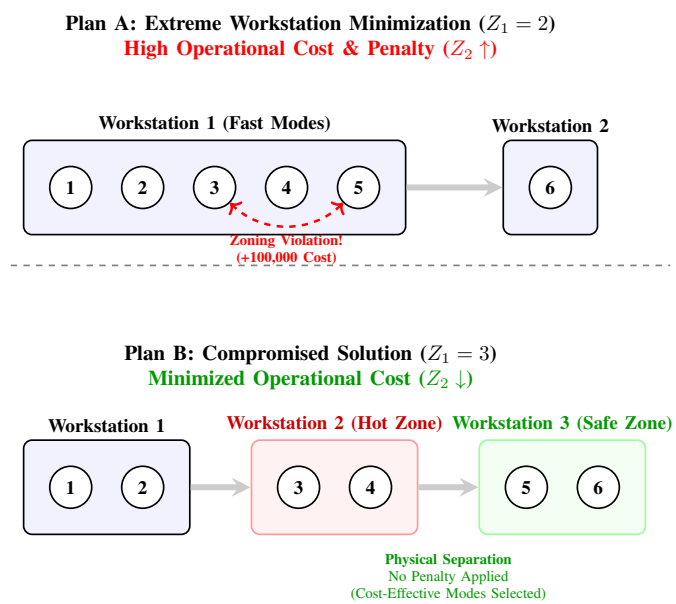

Fig. 3. Schematic layout comparing two assembly line configurations. Plan A forces tasks into fewer workstations, resulting in severe zoning violations and expensive execution modes. Plan B (proposed by Metaheuristic) opens an additional workstation to physically separate incompatible tasks, significantly reducing the total operational cost.

Fig. 3 provides a schematic comparison of two distinct assembly line configurations, illustrating the fundamental trade-off between minimizing the number of workstations (Z_1) and minimizing the total operational cost (Z_2). Plan A demonstrates an extreme workstation minimization strategy, compressing the tasks into only two workstations. However, this dense spatial configuration inevitably forces incompatible tasks (e.g., tasks 3 and 5) to share the same workstation. This triggers a severe soft zoning violation, which heavily penalizes the total operational cost. Furthermore, the restricted space and cycle time compel the selection of fast but highly expensive execution modes.

In contrast, Plan B represents a highly efficient compromised solution typically identified by metaheuristic approaches. By deliberately opening an additional workstation (totaling three workstations), the system effectively isolates the conflicting tasks into physically distinct areas (e.g., separating the Hot Zone from the Safe Zone), successfully completely avoiding the substantial penalty cost. Moreover, this relaxed spatial arrangement provides sufficient time flexibility within the workstations, allowing the algorithms to select slower, cost-effective execution modes. Ultimately, Plan B achieves a significantly lower total operational cost, proving that a slight increase in Z_1 can be strategically justified to optimize Z_2.

Furthermore, the comparative analysis deliberately focuses on the distribution of feasibility rates, specifically highlighting full feasibility, partial feasibility, and complete deadlocks rather than comparing the average values of the objective functions (Z_1 and Z_2). When the rigid model encounters a deadlock (0% feasibility), it mathematically returns arbitrarily large dummy values, such as $99,999,999$ for the total cost. Including these massive outliers would severely distort the statistical averages, rendering any direct comparison of Z_1 and Z_2 mathematically invalid. Therefore, tracking the frequency of deadlocks provides a much more accurate, transparent, and meaningful assessment of the model's structural robustness. A detailed performance comparison between the two models is summarized in Table I.

TABLE I
PERFORMANCE COMPARISON BETWEEN HARD AND SOFT CONSTRAINTS MODELS

Grp.	Model	Feas. (%)	Time (s)	Full	Partial	Deadlock
A	Hard	100.00	0.04	50	0	0
	Soft	100.00	0.08	50	0	0
B	Hard	100.00	3.45	50	0	0
	Soft	100.00	3.53	50	0	0
C	Hard	79.20	13.47	34	10	6
	Soft	**75.60**	15.77	**24**	**21**	**5**

*Note: Each group consists of 50 problem instances.
Full: Fully Feasible (100% adherence to zoning rules).
Partial: Partially Feasible (viable solutions with soft constraint penalties).
Deadlock: Infeasible (0% feasibility, solver failed to find a valid configuration).

Table I summarizes the computational performance and feasibility outcomes of the Hybrid VNS algorithm under both hard and soft constraint models across three problem size categories. As observed in Groups A and B, which represent small and medium-sized instances respectively, both constraint paradigms demonstrate an identical and flawless feasibility rate of 100%. Furthermore, the average computational times required by both approaches remain highly comparable, taking merely fractions of a second for Group A and approximately 3.5 seconds for Group B. This indicates that when the objective search space is relatively small, the strictness of spatial constraints does not significantly impair algorithmic performance or solution discovery.

However, a distinct performance divergence emerges when the algorithms encounter the highly constrained, large-scale instances of Group C. Under the strict hard constraints model, the system experiences a severe computational bottleneck, resulting in 6 complete deadlocks (0% feasibility) where no viable assembly line configuration could be mathematically identified. Conversely, transitioning to the soft constraints framework effectively mitigates this critical failure. By permitting zoning violations in exchange for penalty costs, the soft model successfully salvages previously unsolvable instances, notably increasing the number of partially feasible solutions from 10 to 21 and reducing the total number of deadlocks to 5. Although this relaxation marginally increases the average computational time (15.77 seconds) and shifts the distribution of fully feasible outcomes, it crucially provides decision-makers with actionable, alternative production plans in extreme scenarios where the hard constraint model simply collapses.

C. Experimental Result

This chapter presents the performance evaluation of the implemented algorithmic approaches. Specifically, the Exact Method (CPLEX), Hybrid VNS, Pure VNS, MOEA/D, NSGA-II, and MO-VNS. In solving the Multi-Objective Assembly Line Balancing Problem with Soft Zoning Constraints (MOALBP-SZ).

To systematically evaluate the algorithmic performance across varying degrees of complexity, the benchmark datasets are categorized into three distinct groups based on the total number of tasks (n). Group A consists of small-scale instances where $n < 20$. Group B represents medium-scale instances containing tasks in the range of $20 \leq n \leq 50$. Finally, Group C encompasses highly complex, large-scale instances where $n > 50$.

It is imperative to demonstrate the practical efficacy of the algorithms in optimizing the actual objective functions: minimizing the total number of workstations (Z_1) and minimizing the total penalty costs associated with zoning violations (Z_2). To explicitly illustrate this capability, two representative instances were selected from each group. The following section presents a direct comparison of the best-found objective values (Z_1 and Z_2) achieved by each algorithmic approach on these selected representative problems.

1) Performance Comparison in Small Instances (Group A):

- Comparison of algorithms in optimizing the objective functions

TABLE II
COMPARISON OF ALGORITHMS IN OPTIMIZING THE OBJECTIVE FUNCTIONS: JACKSON (N = 11)

Algorithm	Z_1	Z_2	**Time (s)**
Exact Method	**5**	123	0.9946
Pure VNS	**6**	**95**	**0.0004**
Hybrid VNS	7	**95**	0.0641
NSGA-II	7	**95**	0.3358
MOEA/D	**6**	**102**	0.6846
MO-VNS	**6**	**102**	**0.0110**

* Processing time limit set to 600 seconds/Instance.

Table II evaluates the algorithmic performance on the small-scale Jackson instance (N = 11). While the Exact Method achieves the minimum number of workstations ($Z_1 = 5$), it yields the highest total operational cost ($Z_2 = 123$). Conversely, metaheuristic approaches provide more cost-effective alternatives. For instance, Pure VNS achieves the lowest total cost ($Z_2 = 95$) by utilizing 6 workstations. Notably, MO-VNS identifies the exact same high-quality compromised solution as MOEA/D ($Z_1 = 6$, $Z_2 = 102$) but operates significantly faster, requiring only 0.011 seconds compared to 0.6846 seconds, thereby proving its superior computational efficiency in small-scale scenarios.

TABLE III
COMPARISON OF ALGORITHMS IN OPTIMIZING THE OBJECTIVE FUNCTIONS: MERTENS (N = 7)

Algorithm	Z_1	Z_2	**Time (s)**
Exact Method	2	77	0.1630
Pure VNS	2	63	**0.0002**
Hybrid VNS	2	**56**	0.0692
NSGA-II	2	63	0.3236
MOEA/D	2	63	0.4345
MO-VNS	2	63	**0.0040**

* Processing time limit set to 600 seconds/Instance.

Table III evaluates the algorithmic performance on the Mertens instance (N = 7). For this extremely small-scale problem, all evaluated approaches successfully minimize the number of workstations to an identical optimal value ($Z_1 = 2$). The primary performance divergence is observed in the total operational cost (Z_2). Notably, the Hybrid VNS outperforms all other methods by achieving the lowest total cost ($Z_2 = 56$), demonstrating the effectiveness of the exact solver during the decoding phase in optimizing task assignments within highly restricted search spaces. In contrast, the Exact Method yields the highest cost ($Z_2 = 77$). Meanwhile, the multi-objective algorithms (NSGA-II, MOEA/D, and MO-VNS) consistently converge to a total cost of 63, with MO-VNS remaining computationally efficient by identifying this solution in merely 0.004 seconds.

- Algorithm Performance Evaluation:

  The experimental results for the small problem instances clearly illustrate the distinct characteristics of each algorithmic approach. The computational findings reveal that single-objective oriented algorithms. There are specifically the Exact Method, Hybrid VNS, and Pure VNS. The algorithms yielded relatively low Hypervolume Ratio (HVR) coverage (0.384, 0.584, and 0.552, respectively) and exhibited high Additive Epsilon Indicator (I_ϵ) values. This behavior is a mathematically anticipated outcome. Because these algorithms generate a single extreme point solution primarily focused on minimizing the number of workstations (Z_1), they inherently lack the capacity to comprehensively cover the objective space in the manner of a true Pareto Front.

TABLE IV
PERFORMANCE COMPARISON IN SMALL INSTANCES (GROUP A)

Algorithm	**HVR**	**Epsilon I_ϵ**	**Time (s)**
Exact Method	0.384290	0.900022	0.298408
Pure VNS	0.552214	0.763334	**0.000156**
Hybrid VNS	0.584082	0.730000	0.075466
NSGA-II	0.627478	0.699986	0.329656
MOEA/D	**0.639538**	0.683320	0.539156
MO-VNS	0.636432	**0.683318**	**0.008570**

As detailed in the summary table IV. for Group A, an analysis of the multi-objective metaheuristics demonstrates that MOEA/D achieved the highest solution diversity quality (HVR = 0.639). However, MO-VNS algorithm secured a highly competitive HVR of 0.636 while exhibiting remarkable superiority in computational efficiency. MO-VNS required an average processing time of merely 0.008 seconds, rendering it approximately 67 times faster than MOEA/D, which consumed 0.539 seconds. These results robustly confirm that the local search mechanism of the MO-VNS is highly agile and optimally suited for solving small-scale instances.

2) Dimensional Transition in Medium Instances (Group B):

- Comparison of algorithms in optimizing the objective functions

TABLE V
COMPARISON OF ALGORITHMS IN OPTIMIZING THE OBJECTIVE FUNCTIONS: KILBRID (N = 45)

Algorithm	Z_1	Z_2	**Time (s)**
Exact Method	**8**	556	11.8314
Pure VNS	12	360	**0.0188**
Hybrid VNS	12	360	8.7187
NSGA-II	12	360	0.6228
MOEA/D	12	360	1.6156
MO-VNS	12	360	**0.0465**

* Processing time limit set to 600 seconds/Instance.

Table V evaluates the algorithmic performance on the medium-scale Kilbridge instance (N = 45). The results

clearly illustrate a significant multi-objective trade-off. While the Exact Method minimizes the number of workstations to the absolute minimum ($Z_1 = 8$), it incurs a substantially high total operational cost ($Z_2 = 556$) and requires the longest computational time (11.83 seconds). In contrast, all metaheuristic approaches consistently converge on an identical compromised solution, utilizing 12 workstations to significantly reduce the total cost to 360. Among the multi-objective algorithms achieving this balanced result, MO-VNS demonstrates superior computational efficiency, requiring merely 0.0465 seconds compared to NSGA-II (0.6228 seconds) and MOEA/D (1.6156 seconds).

TABLE VI
COMPARISON OF ALGORITHMS IN OPTIMIZING THE OBJECTIVE FUNCTIONS: ROSZIEG (N = 25)

Algorithm	Z_1	Z_2	**Time (s)**
Exact Method	**4**	284	0.712
Pure VNS	6	**200**	**0.002**
Hybrid VNS	**5**	207	1.399
NSGA-II	6	**200**	0.545
MOEA/D	6	207	1.198
MO-VNS	6	207	**0.032**

* Processing time limit set to 600 seconds/Instance.

Table VI evaluates the algorithmic performance on the medium-scale Roszieg instance (N = 25) The Exact Method strictly minimizes the workstations to 4, which consequently drives the total operational cost to the highest value of 284. In contrast, the Hybrid VNS discovers a unique compromise, balancing the system with 5 workstations and reducing the cost to 207. Meanwhile, Pure VNS and NSGA-II successfully identify the lowest total operational cost of 200 by utilizing 6 workstations. Notably, MO-VNS achieves a highly competitive alternative solution ($Z_1 = 6, Z_2 = 207$) in a mere 0.032 seconds, demonstrating its capability to rapidly generate effective production plans compared to MOEA/D, which requires 1.198 seconds for the identical objective values.

- Algorithm Performance Evaluation: As the problem size expands in Group B, the impact of the exponential growth of the search space becomes tangibly apparent. The computational time of the Exact Method surged from 0.29 seconds in small instances to 5.95 seconds, while the Hybrid VNS required an increased processing time of 3.53 seconds. This phenomenon clearly indicates the inherent limitations of Mixed-Integer Linear Programming (MILP) and Constraint Programming (CP) solvers when the number of decision variables increases significantly.

As demonstrated in the summary table VII, within this broader search space, MOEA/D exhibited the most robust potential in terms of solution quality, achieving the highest HVR score of 0.647 and the lowest Additive Epsilon Indicator (I_ϵ) of 0.671. Meanwhile, NSGA-II yielded a comparable performance (HVR = 0.631) but maintained a competitive advantage in computational speed, requiring only 0.63 seconds (over twice as fast as MOEA/D). Furthermore, MO-VNS algorithm consistently maintained an exceptional balance in speed, utilizing merely 0.045 seconds. This remarkable efficiency establishes MO-VNS as a highly suitable alternative for scenarios where users are subject to strict time constraints.

TABLE VII
PERFORMANCE COMPARISON IN MEDIUM INSTANCES (GROUP B)

Algorithm	**HVR**	**Epsilon I_ϵ**	**Time (s)**
Exact Method	0.338778	0.822018	5.953632
Pure VNS	0.322514	0.892966	**0.007744**
Hybrid VNS	0.528870	0.733500	3.534928
NSGA-II	0.631456	0.672182	0.634206
MOEA/D	**0.647166**	**0.671482**	1.527070
MO-VNS	0.590168	0.709808	**0.045138**

3) Computational Limits and Performance in Large Instances (Group C):

- Comparison of algorithms in optimizing the objective functions

TABLE VIII
COMPARISON OF ALGORITHMS IN OPTIMIZING THE OBJECTIVE FUNCTIONS: ARC111 (N = 111)

Algorithm	Z_1	Z_2	**Time (s)**
Exact Method	**15**	1140	600.850
Pure VNS	22	888	**0.312**
Hybrid VNS	21	888	12.481
NSGA-II	22	888	1.013
MOEA/D	**20**	909	4.112
MO-VNS	22	888	**0.4926**

* Processing time limit set to 600 seconds/Instance.

Table VIII evaluates the algorithmic performance on the large-scale Arc111 instance (N = 111). The results illustrate the severe impact of the exponential growth of the search space; the Exact Method experiences a severe computational bottleneck, exceeding the 600-second time limit. Although it aggressively minimizes the workstations ($Z_1 = 15$), it yields the highest total operational cost ($Z_2 = 1140$). In contrast, the Hybrid VNS identifies an exceptional compromise, achieving the lowest total cost ($Z_2 = 888$) with 21 workstations. Furthermore, MOEA/D successfully reduces the workstations to 20 at a marginally higher cost ($Z_2 = 909$). Meanwhile, Pure VNS, NSGA-II, and MO-VNS converge on an identical high-quality solution ($Z_1 = 22, Z_2 = 888$), with MO-VNS locating this alternative in merely 0.4926 seconds, thereby demonstrating its robust scalability and speed in highly complex environments.

Table IX evaluates the algorithmic performance on the most complex large-scale instance, Barthol2 (N = 148). The Exact Method completely succumbs to the exponential growth of the search space, exceeding

TABLE IX
COMPARISON OF ALGORITHMS IN OPTIMIZING THE OBJECTIVE FUNCTIONS: BARTHOL2 (N = 148)

Algorithm	Z_1	Z_2	Time (s)
Exact Method	**38**	1709	602.253
Pure VNS	54	1191	1.603
Hybrid VNS	54	**1184**	15.264
NSGA-II	54	1219	1.509
MOEA/D	**50**	1275	6.340
MO-VNS	52	1219	**1.168**

* Processing time limit set to 600 seconds/Instance.

the 600-second computational limit (602.253 seconds) and resulting in an excessively high total operational cost ($Z_2 = 1709$) despite severely restricting the workstations to 38. In contrast, the metaheuristics successfully navigate this dense search space. The Hybrid VNS achieves the lowest total operational cost ($Z_2 = 1184$) by utilizing 54 workstations, whereas MOEA/D aggressively minimizes workstations to 50 at a higher cost ($Z_2 = 1275$). Remarkably, MO-VNS strikes a highly practical balance ($Z_1 = 52, Z_2 = 1219$) and records the fastest computational time of a mere 1.168 seconds. This proves the exceptional efficiency and practical viability of MO-VNS in generating robust alternatives for massive industrial assembly lines.

- Algorithm Performance Evaluation: Testing with large-scale datasets (Group C) serves as a crucial validation reflecting the realities of actual industrial environments. The experimental results indicate a computational collapse of the exact methods; the average processing time of the Exact Method surged to 537.18 seconds (an increase of over 1,800 times compared to small instances). In numerous test cases, the system experienced timeouts without successfully confirming the ideal point, subsequently causing the Hypervolume Ratio (HVR) to deteriorate to 0.502. Similarly, the Hybrid VNS encountered a severe computational bottleneck during the decoding phase utilizing the CP Solver, which escalated its processing time to 15.76 seconds.

TABLE X
PERFORMANCE COMPARISON IN LARGE INSTANCES (GROUP C)

Algorithm	HVR	Epsilon I_ϵ	Time (s)
Exact Method	0.502103	0.633636	537.184892
Pure VNS	0.367642	0.805578	**2.600760**
Hybrid VNS	0.436030	0.749286	15.769212
NSGA-II	**0.643394**	**0.529714**	4.400764
MOEA/D	0.601298	0.589360	19.013622
MO-VNS	0.532380	0.620106	**3.056526**

As demonstrated in the summary table X , NSGA-II validated itself as the most robust algorithm by successfully outperforming the vector distribution capabilities of MOEA/D. NSGA-II achieved the highest HVR score of 0.643 and the lowest Additive Epsilon Indicator (I_ϵ) of 0.529, requiring a mere 4.40 seconds of computational time. This success is fundamentally attributed to the efficiency of its non-dominated sorting mechanism integrated with crowding distance evaluation. This powerful combination effectively preserves solution diversity and prevents the clustering of answers in any localized region within the highly intricate search space. Meanwhile, MO-VNS algorithm consistently preserved its prominent advantage as the fastest multi-objective algorithm, utilizing only 3.05 seconds, thereby maintaining its computational superiority consistently across all problem scales. As shown in Fig. 4

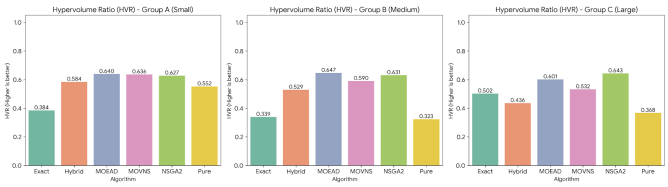

Fig. 4. Hypervolume Comparison

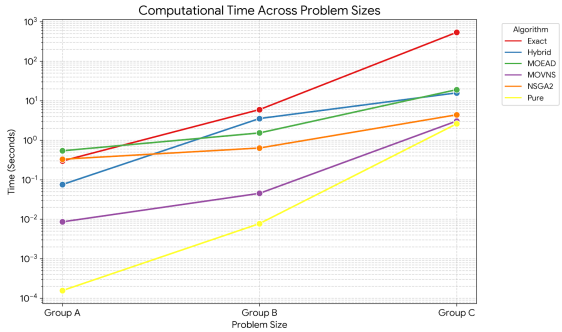

Fig. 5. The comparison of Hypervolume Computational time

As shown in Fig. 5 the comparison of Hypervolume Computational time required by six different algorithms across three escalating problem size categories (Group A, Group B, and Group C). As depicted on the logarithmic y-axis, all evaluated approaches inherently experience a substantial, exponential increase in processing time as the problem dimensions expand. Notably, the Exact Method (indicated by the red line) exhibits the most severe performance degradation; its computational time surges drastically in Group C, demonstrating its susceptibility to the exponential growth of the search space. Similarly, the Hybrid VNS (blue line) displays a steep upward trajectory, reflecting the severe computational bottlenecks encountered during its exact solver-based decoding phase in larger instances. In contrast, the pure metaheuristic algorithms demonstrate significantly superior scalability. While Pure VNS and MO-VNS maintain exceptionally low processing times in small and medium instances, algorithms like NSGA-II and MO-VNS prove to be the most computationally efficient frameworks in the highly complex Group C scenario, suc-

cessfully bypassing the severe computational limitations that incapacitate traditional and hybrid exact solvers.

V. Conclusion

This research presents a solution for the Multi-Objective Assembly Line Balancing Problem with Soft Zoning Constraints (MOALBP-SZ). The primary objectives of this study are to minimize the number of workstations (Z_1) and to minimize the total cost, which incorporates penalty costs incurred from zoning rule violations (Z_2). The study developed and compared the performance of several algorithmic approaches, namely the Exact Method, Hybrid VNS, Pure VNS, MOEA/D, NSGA-II, and MO-VNS, across benchmark datasets of varying scales (small, medium, and large instances).

The empirical results substantiate the fundamental mathematical principle of the "No Free Lunch" theorem, indicating that no single algorithm achieves optimal performance across all scenarios or problem dimensions. The algorithmic behaviors observed in this study can be summarized as follows:

- **Small Instances (Group A):** The Exact Method and Hybrid VNS execute rapidly but yield only a single extreme point solution. In contrast, multi-objective metaheuristics such as MO-VNS can efficiently approximate the Pareto front in mere fractions of a second.
- **Medium Instances (Group B):** As the search space expands, MOEA/D demonstrates the highest potential in solution diversity (measured by Hypervolume), while NSGA-II achieves comparable results at twice the computational speed.
- **Large Instances (Group C):** The exponential growth of the search space causes the Exact Method to fail entirely, resulting in computational timeouts. In this highly complex landscape, NSGA-II emerges as the most robust algorithm, successfully identifying a diverse set of solutions that closely converge toward the ideal Pareto front.

A. Managerial Implications

Based on the experimental findings, this research provides practical guidelines for industrial managers to select the most appropriate algorithms aligned with their organizational scale and context:

- **Small and Medium-sized Enterprises (SMEs):** For facilities with small-to-medium assembly lines and relatively low constraint complexity, the application of Hybrid VNS (or MO-VNS) is highly recommended. These algorithms feature straightforward structures, easily adjustable parameters, and the capability to generate high-quality solutions rapidly, thereby effectively supporting day-to-day production planning.
- **Large-scale Enterprises:** For factories dealing with a massive volume of tasks and highly complex zoning rules (e.g., in the automotive or electronics industries), management should strictly avoid relying on Exact Methods. Instead, it is recommended to integrate NSGA-II or MOEA/D into Enterprise Resource Planning (ERP) systems. These algorithms can maintain system balance and provide multi-objective trade-off alternatives for managerial decision-making within a matter of seconds.

B. Future Work

Although this research successfully integrated soft constraints with metaheuristic approaches, several aspects can be further expanded to enhance industrial realism in future studies:

- **Expanding Task Process Alternatives:** In this study, the benchmark instances defined mode characteristics into only two formats: Fast Mode (requires less time but consumes high resources) and Slow Mode (requires more time but conserves resources). However, within the context of modern industry, production processes exhibit much higher flexibility.

Therefore, future research should develop models that accommodate at least three execution modes. Examples include incorporating a more granular inverse relationship between time and cost, or introducing a Human-Robot Collaboration Mode, which possesses unique characteristics regarding time, cost, and energy consumption. Introducing a third execution mode will not only better reflect the realistic dynamics of a Smart Factory but will also serve as a rigorous stress test for the computational robustness of the NSGA-II and MOEA/D algorithms within an exponentially expanded search space.

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